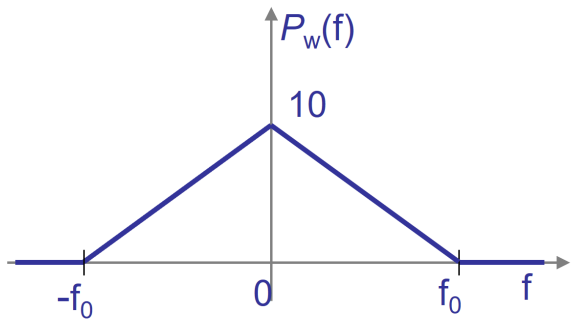
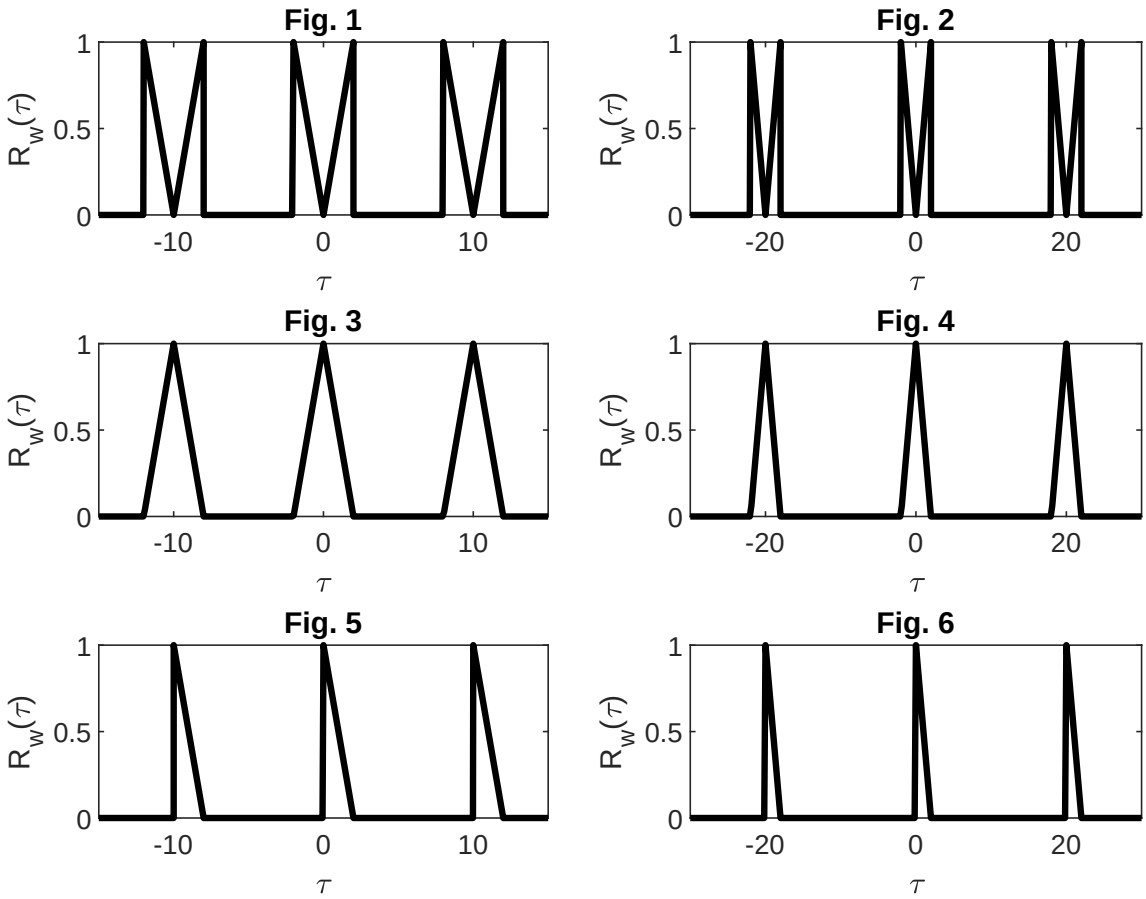
ELE 537, Communication Systems
Exam 1, Fall 2022
Instructor: Dr. Chadi Abou-Rjeily

Problem 1: (5 points)
Consider a signal whose power spectral density is given by $\mathcal{P}_w(f) = 10\text{tri}\left(\frac{f}{f_0}\right)$. Find the value of f_0 if the power contained in the positive frequencies is equal to 100 W.



The area under the triangular function (for positive frequencies) is given by: $\frac{10\times f_0}{2} = 100 \;\Rightarrow\; f_0 =$ 20 Hz.

Problem 2: (5 points)
Consider a periodic signal with period 10 sec. Which one of the below figures constitutes a valid option of the autocorrelation function? Justify your answer.



- The autocorrelation function must be periodic with period 10 sec $\Rightarrow$ Fig. 1, Fig.3 or Fig. 5.
- The autocorrelation function is even $\Rightarrow$ Fig. 1 or Fig.3.
- The autocorrelation function is maximum at $\tau = 0$ $\Rightarrow$ Fig.3.

Therefore, Fig.3 is a valid option of the autocorrelation function.

Problem 3: (5 points)
Are the functions $\cos\left(\frac{\pi}{3}t\right)$ and $\cos\left(\frac{\pi}{6}t\right)$ orthogonal over the time interval $[0 \;\; 12]$ sec?

$$I = \int_0^{12} \cos\left(\frac{\pi}{3}t\right)\cos\left(\frac{\pi}{6}t\right)\mathrm{d}t = \frac{1}{2}\int_0^{12}\cos\left(\frac{\pi}{6}t\right)\mathrm{d}t + \frac{1}{2}\int_0^{12}\cos\left(\frac{\pi}{2}t\right)\mathrm{d}t$$
$$= \frac{1}{2}\underbrace{\int_0^{12}\cos\left(\frac{2\pi}{12}t\right)\mathrm{d}t}_{I_1} + \frac{1}{2}\underbrace{\int_0^{12}\cos\left(\frac{2\pi}{4}t\right)\mathrm{d}t}_{I_2}$$

- $I_1 = 0$ since the duration 12 is a multiple of the period of the cosine that is equal to 12.
- $I_2 = 0$ since the duration 12 is a multiple of the period of the cosine that is equal to 4.

Therefore, $I = 0$ and the functions are orthogonal.

Problem 4: (15 points: $(3,2,2,4,4)$**)**
The signal $x(t) = 1 + 6\sin(4\pi t)$ is applied to the input of a filter whose transfer function is given by $1 + jf$.

a. Find the Fourier Transform of the input signal and deduce its power spectral density (PSD).

b. Deduce the autocorrelation function.

c. Check for the general properties of the autocorrelation function.

d. Find the percentage of the total power of $x(t)$ that falls in the frequency interval $[0.5 \;\; 3]$ Hz.

e. Find the PSD of the output signal.

a.

$$X(f) = \delta(f) - 3j\delta(f-2) + 3j\delta(f+2) \quad \Rightarrow \quad \mathcal{P}_x(f) = \delta(f) + 9\delta(f-2) + 9\delta(f+2)$$

b.

$$R_x(\tau) = \mathfrak{F}^{-1}[\mathcal{P}_x(f)] = 1 + 18\cos(4\pi\tau)$$

c. The autocorrelation function is real, even, periodic with period 0.5 sec and $\max|R_x(\tau)| = R_x(0)$.

d.

$$\begin{aligned} P_{[0.5\ 3]} &= \int_{0.5}^{3} \mathcal{P}_x(f)\mathrm{d}f \\ &= \underbrace{\int_{0.5}^{3} \delta(f)\mathrm{d}f}_{=0} + 9\underbrace{\int_{0.5}^{3} \delta(f-2)\mathrm{d}f}_{=1} + 9\underbrace{\int_{0.5}^{3} \delta(f+2)\mathrm{d}f}_{=0} = 9 \ \ (\mathrm{W}) \end{aligned}$$

$$P_t = R_x(0) = 19 \ \ (\mathrm{W}) \quad \Rightarrow \quad \text{Percentage Power} = \frac{9}{19} \times 100 = 47.36\%$$

e.

$$\begin{aligned} \mathcal{P}_y(f) &= |H(f)|^2 \mathcal{P}_x(f) = (1+f^2)\mathcal{P}_x(f) \\ &= (1+f^2)_{f=0}\delta(f) + 9(1+f^2)_{f=2}\delta(f-2) + 9(1+f^2)_{f=-2}\delta(f+2) \\ &= \delta(f) + 45\delta(f-2) + 45\delta(f+2) \end{aligned}$$

Problem 5: (10 points: $(4,3,3)$**)**

a. Find the 3-dB bandwidth of a signal whose power spectral density is given by $e^{-\frac{f^2}{100}}$.

b. Find the absolute bandwidth of the signal $x(t) = \text{sinc}^2(20t)$.

c. Find the null bandwidth of the signal $x(t) = \text{rect}(100t)$.

a.

$$e^{-\frac{f^2}{100}} = \frac{1}{2} \quad \Rightarrow \quad f = B_{3dB} = \sqrt{100 \times \ln(2)} = 8.32 \ \ \text{(Hz)}$$

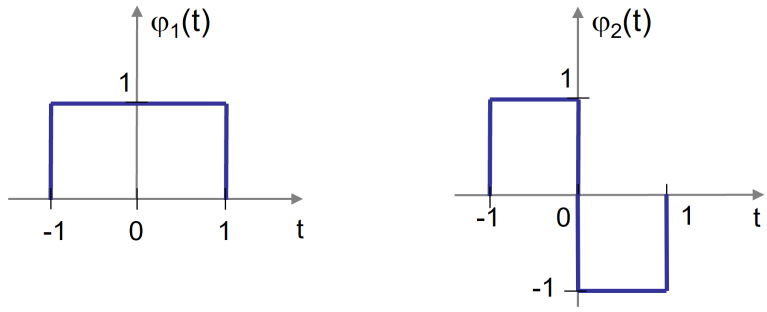
b.

$$X(f) = \frac{1}{20}\text{tri}\left(\frac{f}{20}\right) \quad \Rightarrow \quad B_{abs} = 20 \ \ \text{(Hz)}$$

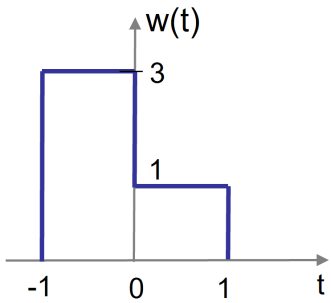
c.

$$X(f) = \frac{1}{100}\text{sinc}\left(\frac{f}{100}\right) \quad \Rightarrow \quad B_{null} = 100 \ \ \text{(Hz)}$$

Problem 6: (10 points: $(7,3)$**)**
Consider the orthogonal functions $\phi_1(t)$ and $\phi_2(t)$ shown in the figure below:



Consider the signal $w(t)$ shown in the figure below:



a. Expand the waveform $w(t)$ into an orthogonal series using the orthogonal set $\{\phi_1(t), \phi_2(t)\}$.

b. Evaluate the mean square error for the orthogonal series obtained in part (a). Is the orthogonal set $\{\phi_1(t), \phi_2(t)\}$ complete?

a. $K_1 = \int_{-1}^{1} \phi_1^2(t)\mathrm{d}t = \int_{-1}^{1} \mathrm{d}t = 2$ and $K_2 = \int_{-1}^{1} \phi_2^2(t)\mathrm{d}t = \int_{-1}^{1} \mathrm{d}t = 2$.

$$a_1 = \frac{1}{K_1}\int_{-1}^{1} w(t)\phi_1^*(t)\mathrm{d}t = \frac{1}{2}\left[\int_{-1}^{0} 3\mathrm{d}t + \int_{0}^{1} \mathrm{d}t\right] = 2$$
$$a_2 = \frac{1}{K_2}\int_{-1}^{1} w(t)\phi_2^*(t)\mathrm{d}t = \frac{1}{2}\left[\int_{-1}^{0} 3\mathrm{d}t - \int_{0}^{1} \mathrm{d}t\right] = 1$$

implying that the orthogonal series representation of $w(t)$ is given by $2\phi_1(t) + \phi_2(t)$.

b. Since $w(t) = 2\phi_1(t) + \phi_2(t)$, then $MSE = 0$ implying that the set $\{\phi_1(t), \phi_2(t)\}$ is complete.

Problem 7: (5 points)
Consider the waveform $w(t)$ that is bandlimited to 20 Hz and let:

$$I = \int_{-\infty}^{+\infty} w(t)\mathrm{sinc}(100(t-2))\mathrm{d}t$$

Show that I can be written as $I = kw(t_0)$ where k and t_0 are two constants to be determined (*hint*: apply Parseval's theorem and the definition of the inverse Fourier transform).

$$\begin{aligned}
I &= \int_{-\infty}^{+\infty} w(t)\mathrm{sinc}(100(t-2))\mathrm{d}t \\
&= \int_{-\infty}^{+\infty} w(t)\left[\mathrm{sinc}(100(t-2))\right]^* \mathrm{d}t \\
&= \int_{-\infty}^{+\infty} W(f)\left[\frac{1}{100}\mathrm{rect}\left(\frac{f}{100}\right)e^{-j4\pi f}\right]^* \mathrm{d}f \quad \text{(Parseval's theorem)} \\
&= \frac{1}{100}\int_{-50}^{+50} W(f)e^{j4\pi f}\mathrm{d}f \\
&= \frac{1}{100}\underbrace{\int_{-\infty}^{+\infty} W(f)e^{j4\pi f}\mathrm{d}f}_{=w(2)\quad \text{definition of the IFT}} \quad \text{(the spectrum of } w(t) \text{ does not exceed 20 Hz)} \\
&= \frac{1}{100}w(2)
\end{aligned}$$

Therefore, $k = 1/100$ and $t_0 = 2$.

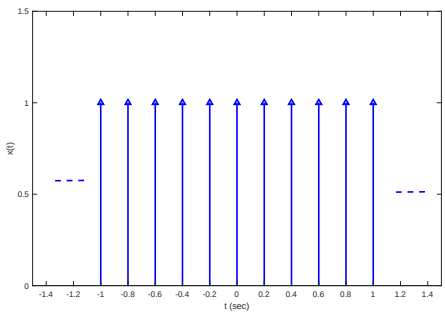
Problem 8: (10 points: $(2,6,2)$**)**
Consider the signal $x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - 0.2n)$.

a. Provide a labeled sketch of the signal.

b. Find the power spectral density (PSD).

c. Provide a labeled sketch of the PSD.

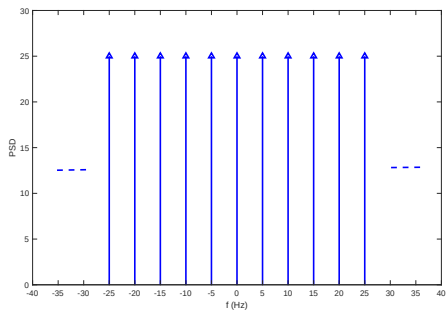
a. $x(t)$ is shown below:



b. Since $x(t)$ is periodic with period $T_0 = 0.2$ sec, then $\mathcal{P}_x(f) = \sum_{n=-\infty}^{+\infty} |c_n|^2 \delta(f - nf_0)$ where $f_0 = 1/0.2 = 5$ Hz.

$$
\begin{aligned}
c_n &= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-j2\pi n f_0 t} \mathrm{d}t \\
&= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \delta(t) e^{-j2\pi n f_0 t} \mathrm{d}t \quad (\text{since } x(t) = \delta(t) \text{ for } t \in [-\frac{T_0}{2} \ \ \frac{T_0}{2}]) \\
&= \frac{1}{T_0} \underbrace{\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \delta(t) \mathrm{d}t}_{=1} = \frac{1}{T_0} = 5 \quad \Rightarrow \quad \mathcal{P}_x(f) = 25 \sum_{n=-\infty}^{+\infty} \delta(f - 5n)
\end{aligned}
$$

c. $\mathcal{P}_x(f)$ is shown below:



Problem 9: (5 points: $(2,3)$**)**
The nonlinear characteristics of the A-law are approximated by piecewise linear chords as follows:

- 4 chords are used for the positive part of the signal.
- Each chord is approximated by a uniform quantizer with 4 steps.
- The step size in chord n is $2^{n-1}\Delta$ for $n = 1,\ldots,4$.
- The same procedure is applied on the negative part of the signal.

a. Find the number of bits per PCM codeword.

b. Find the value of Δ if the analog waveform is limited to the interval $[-12 \;\; 12]$ volts.

a. Each codeword contains 1 sign bit, 2 chord bits and 2 step bits resulting in a total of $n = 5$ bits.

b. Considering the positive part of the signal:

$$4\Delta + 4(2\Delta) + 4(4\Delta) + 4(8\Delta) = 12 \quad \Rightarrow \quad \Delta = \frac{12}{4(1+2+4+8)} = \frac{1}{5} \;\; \text{(V)}$$

Problem 10: (10 points: $(2,4,2,2)$**)**
Consider an analog waveform having an absolute bandwidth of 3 kHz and an amplitude range of -5 V to +5 V. The information in this analog waveform is to be transmitted over a PCM system with the quantizing noise not exceeding $\pm 0.05\%$ of the peak-to-peak signal level.

a. Determine the minimum sampling rate needed.

b. Determine the number of bits needed in each PCM codeword.

c. Determine the minimum bit rate required in the PCM signal.

d. Determine the minimum absolute channel bandwidth required for the transmission of this PCM signal.

a. $f_s \geq 2B = 2 \times 3 = 6$ kHz implying that the minimum sampling rate is equal to 6 kHz.

b. For the error not to exceed $\pm 0.05\%$ of the peak-to-peak value:

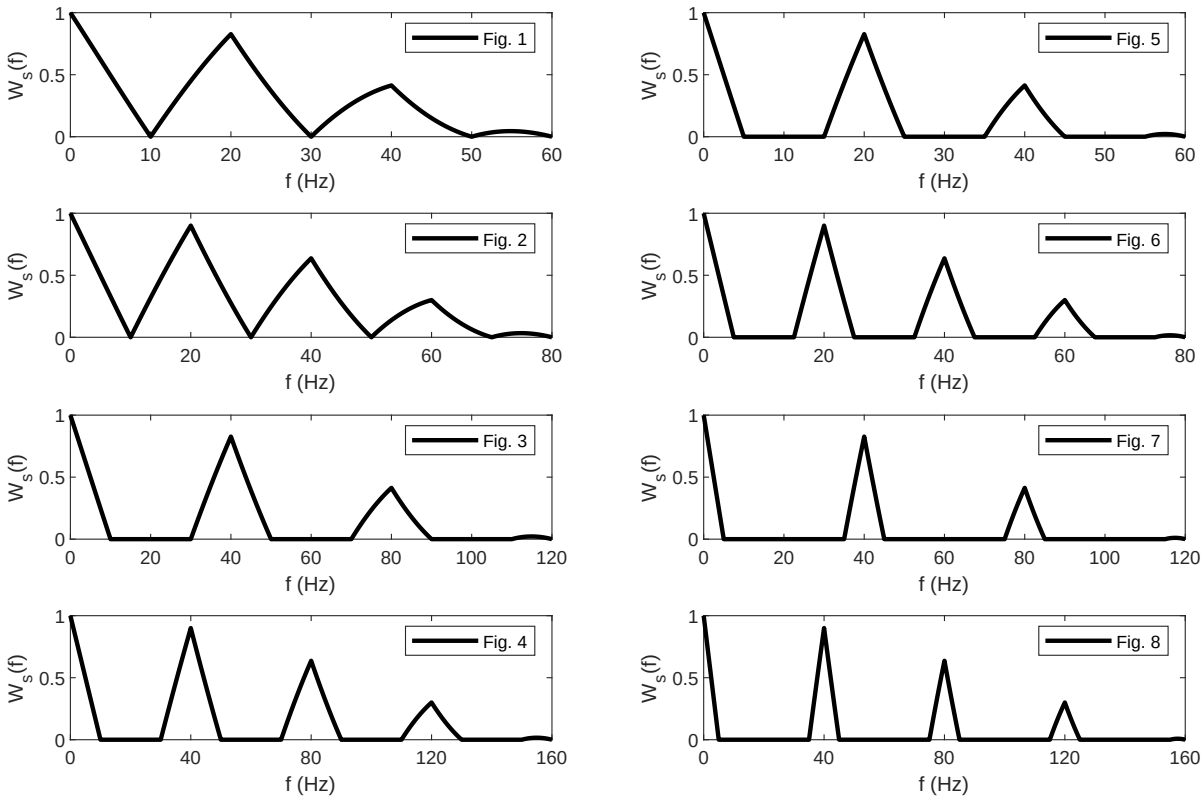
$$\frac{\Delta}{2} \leq \frac{0.05}{100} \underbrace{V_{pp}}_{=2^n\Delta} \quad \Rightarrow \quad 2^n \geq 1000 \quad \Rightarrow \quad n \geq \log_2(1000) \approx 9.93$$

implying that the minimum number of bits is $n = 10$ bits.

c. $R = nf_s = 10 \times 6 = 60$ kbits/s.

d. $B_{min} = \frac{1}{2}R = 30$ kHz.

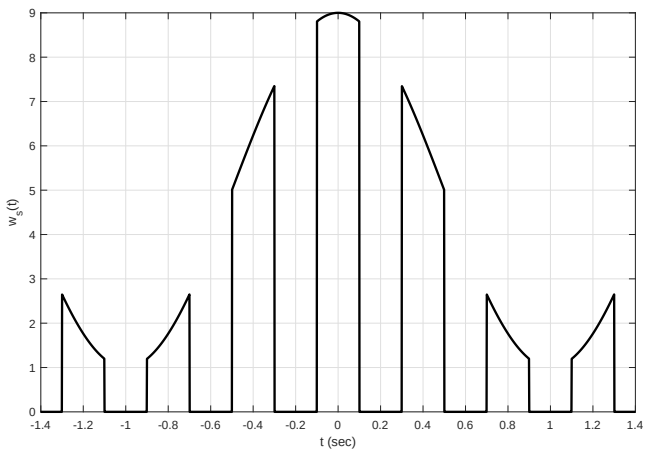
Problem 11: (5 points)
The signal $w(t) = \text{sinc}^2(10t)$ is sampled at the Nyquist rate. Instantaneous PAM is applied with a duty cycle of $1/4$. Which one of the below figures best describes the variations of the spectrum of the sampled signal within its bandwidth? Justify your answer.



$w(t) = \text{sinc}^2(10t) \;\Rightarrow\; W(f) = \frac{1}{10}\text{tri}\left(\frac{f}{10}\right)$ implying that $B = 10$ Hz and $f_s = 2B = 20$ Hz.

- $f_s = 20$ Hz implies that the correct figure is among Fig. 1, Fig. 2, Fig. 5 and Fig. 6.
- $B = 10$ Hz implies that the correct figure is among Fig. 1 and Fig. 2.
- Since $d = 1/4$, then the fourth replica of the spectrum encompasses the zero crossing implying that the correct figure is Fig. 2 ($B_{PAM} = \frac{f_s}{d} = 80$ Hz).

Problem 12: (15 points: $(2,4,2,7)$**)**
Pulse Amplitude Modulation (PAM) is applied on the waveform $w(t) = 5 + 4\cos(\pi t)$ where the sampled signal $w_s(t)$ is shown below.

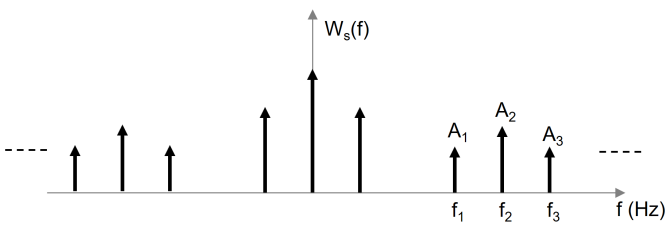


a. Determine the type of sampling (natural or instantaneous).

b. Find the value of the sampling frequency. Will aliasing occur? Justify your answer.

c. Find the value of the duty cycle.

d. If the spectrum $W_s(f)$ of the sampled signal takes the shape shown below, find the values of the frequencies f_1, f_2 and f_3 as well as the amplitudes A_1, A_2 and A_3.



a. Natural PAM since the amplitudes of the rectangular pulses follow the variations of the signal.

b. The pulses are located at multiples of 0.4 implying that $T_s = 0.4$ sec and $f_s = 2.5$ Hz. The signal $w(t)$ has a bandwidth of $f_0 = 0.5$ Hz implying that $f_{Nyq} = 2f_0 = 1$ Hz. Since $f_s \geq 2B$, then aliasing will not occur.

c. From the figure, the width of the pulses is equal to $\tau = 0.2$ sec implying that $d = \frac{\tau}{T_s} = \frac{1}{2}$.

d. $W_s(f) = d\sum_{n=\infty}^{+\infty} \text{sinc}(nd)W(f - nf_s)$ where $f_s = 2.5$ Hz, $d = 1/2$ and:

$$W(f) = 5\delta(f) + 2\delta(f - 0.5) + 2\delta(f + 0.5)$$

The impulses at f_1, f_2 and f_3 correspond to the replica of the spectrum at $n = 1$ (shifted by f_s). Therefore:

$$f_2 = f_s = 2.5 \text{ (Hz)} \quad ; \quad A_2 = 5 \times \underbrace{d}_{=\frac{1}{2}} \underbrace{\text{sinc}(d)}_{=\text{sinc}(1/2)=\frac{\sin(\pi/2)}{\pi/2}=\frac{2}{\pi}} = \frac{5}{\pi}$$

$$f_1 = f_s - 0.5 = 2 \text{ (Hz)} \quad ; \quad A_1 = 2 \times d\text{sinc}(d) = 2 \times \frac{1}{2} \times \frac{2}{\pi} = \frac{2}{\pi}$$

$$f_3 = f_s + 0.5 = 3 \text{ (Hz)} \quad ; \quad A_3 = 2 \times d\text{sinc}(d) = 2 \times \frac{1}{2} \times \frac{2}{\pi} = \frac{2}{\pi}$$